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RESEARCH ARTICLE

Adaptive Credit Scoring Model With Concept Drift Detection and Adaptation Technique for a Dynamic Environment

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ABSTRACT Despite significant advances in Machine Learning (ML) for Credit Scoring, the persistence of drift effects manifested as performance degradation due to evolving borrower behaviour, economic conditions, and shifting data distributions remains a critical challenge. As socioeconomic conditions change, previously learned relationships between input features and target variables become unstable, leading to concept drift, population drift, and label drift. These dynamics reduce model accuracy and reliability, increase misclassification rates, and expose financial institutions to substantial financial and regulatory risks. Traditional static Credit Scoring models lack mechanisms for drift detection and adaptation, while existing adaptive approaches are largely developed for advanced economies and are not well-suited to developing contexts. This study proposes an adaptive credit scoring framework, termed Adaptive Fusion, designed to address drift in developing economy environments. The framework integrates baseline ML models with dynamic adaptation strategies, including retraining, windowing, ensemble learning, and adaptive fusion. Experimental results demonstrate that the Retrained Random Forest, Ensemble, and Adaptive Fusion models achieve superior and consistent performance, each attaining an accuracy of 95.0%, a precision of 0.9275, a recall of 0.9645, an F1-score of 0.9426, and ROC-AUC values exceeding 0.96. Compared to state-of-the-art models such as the Deep Genetic Hierarchical Network of Learners (DGHNL), which achieves 94.60% accuracy and an AUC of 0.9360, the proposed approach demonstrates improved predictive capability, highlighting the effectiveness of dynamic model integration. The Adaptive Fusion algorithm, which dynamically weights and integrates model outputs in real time, emerges as the most robust solution, enabling continuous adaptation to evolving data patterns. These findings underscore the importance of adaptive, drift-aware frameworks for maintaining accuracy, fairness, and reliability in Credit Scoring systems operating in dynamic and resource-constrained environments.

INDEX TERMS Adaptive model, concept drift, credit scoring, machine learning, prediction.

I. INTRODUCTION

Credit scoring estimates the creditworthiness of an individual, a corporation, or even a national entity. Credit scores are calculated based on financial history, current assets, and liabilities. Typically, it tells a lender or investor the probability of the subject being able to pay back a loan [1].

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It is paramount that an entity knows its credit standing. The score acts as a guide on the character of entities; it provides a better description of the creditworthiness of an individual or organization because it combines different attributes. Having the name of an entity is not sufficient to guide one into having a relationship with it. The idea of credit scoring is very noble, as it would give a better description of entities before committing to them [1].

The credit scoring systems in different countries are managed by credit scoring bureaus. The most common scoring categorization uses a range of values and letters in the alphabet. If values are used, the smaller the value, the better the credit standing. Letters of the alphabet range from AAA being excellent to D being poor [2]. The average credit score for a country does not mean that all individuals and organizations within the country have poor credit standing; hence, there is a need for a credit scoring system that shows individual credit scores for people in a country.

Scoring can be performed for a new applicant to estimate credit risk. Behavioral scoring is based on the client's previous or current credit standing; it is believed that the way an individual handles the previous or current loan has a direct link to their future behavior. Collection scoring is used to categorize clients into groups depending on their behavior. Fraud detection classifies applicants according to their probability of being guilty of fraud [3].

The financial environment is so unstable that the data sample used to come up with models might not yield accurate results when the model is used some years after its implementation. Economic conditions in some countries are dynamic, ranging from multi-currency systems to general population drifts. Sometimes the economic conditions improve; in some cases, they degrade, which can contribute to the inaccuracy of static credit scoring models implemented before the changes [3]. This research aims to develop a two-stage credit scoring system that enables the management of the temporal degradation of credit scoring models in general.

Most of the existing traditional systems that are one-shot, fixed-memory-based, and trained from fixed training datasets and static models cannot manage and process highly evolving financial data. Changes that occur within a population cause a change in the distribution of variables; the change in variables would then cause a change in the performance of the model. Hence, there is a need to design an adaptive credit-scoring model for this dynamic environment.

The need for an adaptive credit scoring system is a result of dynamic socioeconomic changes in the environment, emanating from changes in data distribution, such as concept drift [2].

The model has practical applicability across multiple domains. In the insurance industry, it can be utilized to support decision-making processes related to the approval of new policy applications and the renewal of existing policies by assessing the creditworthiness of applicants. In the real estate sector, landlords may employ credit scoring mechanisms to evaluate the likelihood of prospective tenants meeting rental payment obligations in a timely manner. Furthermore, in the field of human resource management, employers may incorporate credit history and credit scores into their recruitment processes, particularly for positions involving the management of substantial financial resources, as an additional measure of assessing candidate reliability and financial responsibility.

The remainder of this paper is organized as follows. In Section II, we provide the research method. In Section III, we outline drift detection, drift understanding, and model adaptation techniques. Section IV outlines the design of the proposed algorithm. In Section V, we present the results, and Section VI provides the conclusion and future work.

II. RESEARCH METHOD

In credit scoring systems, adapting to concept drift, evolving patterns in data over time, is crucial for preserving model accuracy. To address this problem, a range of techniques is employed, such as window-based approaches, drift detection algorithms, and ensemble methods that dynamically adjust to changing data. These adaptation strategies fall into three main categories: instance-based, model-based, and feature-based, each designed to handle both gradual and sudden shifts in the data streams. Recent research has incorporated meta-learning and reinforcement learning to anticipate and respond to drift more effectively, striking a balance between timely detection and robust adaptation.

A. INSTANCE BASED DRIFT ADAPTATION TECHNIQUES

Instance-based approaches leverage the weighting, selection, or transfer learning of samples to counteract drift by emphasizing relevant or recent data instances. Techniques such as importance weighting, local region competence, and instance transfer improve model responsiveness to population shifts and data scarcity in credit scoring, achieving notable Area Under the Curve (AUC) improvements and operational efficiency.

A Diverse Instance-Weighting Ensemble (DiwE) is an instance-based ensemble learning algorithm developed to tackle concept drift in evolving data streams. This concept was first introduced in [4]. Recognizing that concept drift is an inherent property of such streams and that maintaining ensemble diversity is a significant challenge, their work proposed a novel diversity measurement. Unlike traditional methods that assess diversity via inputs, outputs, or classifier parameters, DiwE's approach is based on the degree to which ensemble members disagree with the probability of a regional distribution change. This method uses estimations of regional distribution changes, as instance weights and diversity are fostered by constructing different regional sets through various schemes, leading to varied drift estimation results. To maximize diversity, the algorithm strategically selects classifiers that exhibit the highest disagreement. Evaluations conducted on a range of synthetic and real-world data-stream benchmarks demonstrated the effectiveness and advantages of the proposed DiwE algorithm in adapting to concept drift.

Work done in [5] introduced a novel adaptive behavioural credit scoring scheme designed to address critical issues in credit risk assessment, primarily population in some cases concept drift, which causes underlying data distributions to change over time and necessitates continuous model monitoring and recalibration. The proposed approach utilizes online training for each incoming borrower inquiry by dynamically identifying a specific region of competence using the k-nearest neighbour (kNN) algorithm, which then serves as a localized training set for a tailored classification model. This study compared various classifiers, including logistic regression (LR), random forests (RF), and gradient boosting trees (XGBoost). Key findings indicate that local logistic regression models consistently and statistically significantly outperform their global counterparts and achieve comparable results to more complex global Machine Learning models such as RF and XGBoost. This research also highlights that the selection of a region of competence based on similar characteristics, as opposed to random selection, is crucial for superior performance. The study concluded that this adaptive scheme provides an effective solution for population drift, offering competitive performance while maintaining interpretability, particularly with local LR, which is beneficial for regulatory compliance.

In recent advances in credit risk modelling, an innovative personal credit risk assessment framework based on Instance-Based Transfer Learning (TL) that addresses key limitations in traditional models, such as data sparsity and platform-specific variability, was introduced. The approach rebalances sample weights in the source domain and transfers knowledge to target domains with limited data, thereby uncovering shared patterns across financial platforms [6]. Through integrating base learners such as decision trees, logistic regression, and XGBoost, and applying the model to datasets from P2P platforms and banks, the study demonstrated a significant 24% improvement in AUC compared to conventional Machine Learning techniques. This underscores the robustness and practical applicability of the model in diverse financial contexts. Wang and Yang [6] further emphasized the potential for broader domain adaptation and continued refinement to enhance the model's scalability and effectiveness in real-world credit evaluation scenarios.

The cold-start problem in credit scoring is addressed by introducing transfer-learning techniques tailored to the financial data context [7]. Wei et al. proposed two novel models: the SPY-Transfer model, which adapts the SPY algorithm from positive-unlabelled learning to select valuable source samples for target domain enhancement, and the SPY-TrAdaBoost model, which combines SPY's sample selection with TrAdaBoost's weighting mechanism to improve predictive accuracy. The experimental results demonstrate that both models outperform traditional cold-start problem-solving approaches and the classic TrAdaBoost algorithm, highlighting the effectiveness of sample-based transfer learning in credit scoring tasks.

There is a need to explore the limitations of traditional batch Machine Learning models in dynamic credit scoring environments and advocate the use of data-stream classification techniques. By analysing three anonymized real-world datasets from Brazilian financial institutions, Barddal et al. [8] demonstrated that customer behaviour and feature relevance can drift over time, rendering static models obsolete. They compared batch and stream-based algorithms using the Kolmogorov–Smirnov and population stability index metrics, showing that data stream learners, particularly Adaptive Random Forests, can match or outperform conventional models while maintaining stability.

DriftLens, an unsupervised real-time concept drift detection framework for deep learning models that operates on unstructured data, addresses challenges in monitoring model performance as data distributions change over time [9]. The framework computes the distribution distances of deep learning embedding representations to detect and characterize drift. DriftLens can classify the presence of drift in less than 0.2 seconds, making it at least five times faster than other detectors. It outperformed techniques in 11 out of 13 use cases, demonstrating its high effectiveness in drift discrimination. The correlation index of the drift curve modelled by DriftLens was ≥ 0.85, indicating a strong alignment with the actual drift patterns. DriftLens is robust to parameter settings and ensures consistent performance across various data types and classification tasks. This study addressed the challenges of concept drift detection in dynamic environments, in which actual labels are often unavailable.

A novel approach called CatSight for detecting concept drift in multivariate time series data, particularly in industrial processes, by utilizing Common Spatial Patterns and Machine Learning algorithms, was presented in [10]. The CatSight method addresses concept drift in industrial monitoring by utilizing Common Spatial Patterns to differentiate data distributions and employing machine-learning algorithms for detection. This approach effectively identifies sudden changes in data streams and enhances predictive performance in real-time applications. The evaluation of six state-of-the-art classifiers demonstrated its adequacy. CatSight combines a Common Spatial Pattern (CSP) for feature selection and conventional Machine Learning classifiers for concept drift detection.

The method was tested on three datasets, including two public and one industrial dataset, and showed superior performance. CatSight achieved an average accuracy increase of 10.5% compared to conventional classification methods. The best-performing combination was CSP with Support Vector Machine (SVM), which yielded higher accuracy scores than the other methods. This study emphasized the potential application of CatSight in various industrial problems owing to its effective drift detection capability. Future research should explore classifier ensembles and streaming data analysis to enhance the effectiveness of the method. The study in [10] also discussed the need to address limitations and consider unbalanced class data problems for better drift detection. Overall, CatSight demonstrated improved discrimination rates in multivariate time-series data that were affected by concept drift.

B. MODEL-BASED DRIFT ADAPTATION TECHNIQUES

Model-based methods focus on updating model parameters or structures to accommodate drift, including incremental learning, adaptive ensemble models, and reinforcement learning-driven update scheduling. These strategies enhance predictive performance and computational efficiency but face challenges in balancing the update frequency and avoiding catastrophic forgetting.

Adaptive Model Updating using a Simulated Environment (AMUSE), a reinforcement learning-based framework designed to address concept drift in predictive modelling by learning optimal update policies within a simulated environment, was proposed in [11]. Unlike traditional approaches that rely on fixed schedules or reactive drift detection, AMUSE leverages a parametric model to simulate diverse drift scenarios and trains a model that proactively recommends model updates based on the expected performance gains and update costs. This method enables classifiers to maintain accuracy while minimizing unnecessary retraining, making it particularly valuable in domains such as finance and healthcare, where data distribution evolves and update costs are high. The empirical results demonstrate that AMUSE outperforms conventional strategies in simulated settings, offering a robust and cost-sensitive solution for adaptive model maintenance.

The study in [12] addressed the challenge of concept drift in Machine Learning, particularly in incremental learning scenarios, where data distributions are not static, and training data may be incomplete. Traditional retraining methods are often time-consuming and computationally expensive. To overcome this, they proposed an incremental Support Vector Machine (SVM) learning approach that incorporates domain adaptation. Their method focuses on two critical issues: acquiring new knowledge without forgetting previously learned information, and discarding obsolete data without corrupting valid information. The proposed solution involves fine-tuning the previous model with a small amount of new data to create a model that is sensitive to new information, while retaining old knowledge through parameter transfer. Furthermore, an ensemble and model selection mechanism based on Bayesian theory was introduced to preserve valid information. Experimental results demonstrated improved performance as new data were acquired, and the effectiveness of the algorithm was also explored in relation to the degree of data drift. The model demonstrated performance gains over the standard SVM and incremental SVM algorithms on four out of five industrial datasets and four synthetic datasets.

To address the challenge of credit scoring in scenarios where customer behaviour and financial variables drift over time, an Adaptive and Dynamic Heterogeneous Ensemble (ADHE) approach was introduced in this study. ADHE leverages data stream learning techniques to enable incremental learning and adaptation to drifting variables. The ADHE is designed to exploit diversity by incorporating models derived from various learning algorithms. The model's predictive performance was evaluated using publicly available datasets, and comparisons with existing state-of-the-art models demonstrated that ADHE significantly outperforms them in prediction accuracy, as confirmed by non-parametric tests, while also considering the computational cost [13].

The study in [14] addressed the challenges of credit card fraud detection, particularly the issue of concept drift, where customer spending behaviours and market purchasing trends frequently vary over time, and the inherent imbalanced class distribution in real transaction data. To address these problems, they proposed and investigated a card-based incremental Gradient Boosting Tree (GBT) model. This approach allows the GBT model to incrementally learn from transactions of fraudulent credit cards reported daily, thereby adapting in real time to drifts in online transactions. This study compares card-based incremental GBT with a regular GBT model and a retraining method that combines previous and new fraudulent transaction sets. Experiments conducted on four months of real transaction data (December 2019 to March 2020) demonstrated improvements in fraud detection performance across all months, noting a significant concept drift in December that impacted the GBT model's performance. The effectiveness of the card-based incremental learning was further validated by comparing it with transaction-based incremental learning, which is prone to catastrophic forgetting.

C. FEATURE BASED DRIFT ADAPTATION TECHNIQUES

Feature-based adaptation addresses changes in the feature relevance or distribution through adaptive feature selection, representation learning, and domain adaptation. Approaches include random subspace methods, feature weighting, and transfer learning to mitigate feature drift and redundancy and to enhance model generalization and interpretability in credit risk evaluation.

A recent literature survey by Rabash et al. [15] addressed the significant challenges in stream data learning, particularly focusing on the dynamic behaviours and changes in the environment that lead to concept and feature drifts. This study provides a comprehensive overview of the fundamental concepts and definitions in this domain, along with various models and methods developed for detecting feature drift and maintaining the validity of Machine Learning models when such drifts occur. It details the generators utilized for creating datasets with feature drift, which is crucial for benchmarking detection and handling approaches. The authors also presented a taxonomy of feature selection methods applicable to both static and dynamic environments. Concluding their analysis, they suggested that reinforcement-based models hold significant promise in addressing these challenges.

D. META LEARNING AND SELF-ADAPTIVE SYSTEMS

Meta-learning frameworks, including fairness-aware online meta-learning and lifelong self-adaptation, are advancing credit scoring by dynamically selecting models and adapting them to non-stationary environments. These methods improve robustness against heterogeneous drift patterns and evolving fairness constraints by using bi-level optimization and lifelong ML layers to track the system and environmental changes. They offer promising avenues for continuous learning and decision-making enhancements under drift conditions.

A Concept Drift Detection Framework with Hybrid Meta-Learning (CDDF-HML), a unified framework designed to address the challenge of concept drift in dynamic data stream mining, was introduced in [16]. Recognizing that traditional methods often specialize in either gradual or abrupt drifts, CDDF-HML employs a novel approach that combines meta-learning, adaptive feature selection, and ensemble-based processes to effectively identify both types of concept changes. The framework is particularly suited for environments in which data distributions are continuously evolving, demonstrating its capability to detect deviations and adapt to various data conditions. The comparative analysis performed in this study indicated that CDDF-HML is an effective tool for discovering concept drift, thereby enhancing the reliability of Machine Learning models in dynamic data situations. This work can be further improved by implementing the method in specific domains, refining adjustment approaches, and improving scalability; hence, the purpose of this study is to apply it to Credit Scoring.

The significant challenge of concept drift in data streams, where evolving data characteristics can render classification models obsolete, was addressed in [17]. A core difficulty lies in selecting an appropriate drift detector for a given stream, as prior knowledge may be unavailable, and stream properties may change over time. To overcome this problem, a novel framework that leverages meta-learning was proposed. This framework extracts statistical and temporal meta-features from sliding windows to dynamically recommend the most suitable drift detector in real-time for unseen data chunks based on their properties. The effectiveness of this approach was evaluated through experiments on 10 real-world and 18 synthetic data streams, all of which were subject to concept drift and class imbalance. The results demonstrate that the proposed framework significantly enhances concept drift detection across various scenarios, exhibiting robustness to class imbalance and highlighting the benefits of dynamically adapting the drift detector.

There is a need for a generic approach to learning under concept drift that extends traditional Machine Learning workflows to accommodate dynamic data environments. Instead of the usual two-phase process of training and testing. A generic approach to learning under concept drift that extends traditional Machine Learning workflows to accommodate dynamic data environments was articulated in [18]. The authors outlined a three-phase structure, namely drift detection, drift understanding, and drift adaptation. Drift detection involves identifying changes in the data distribution using various statistical or model-based techniques. Drift understanding focuses on pinpointing when and where a drift occurs, which is crucial for timely and accurate adaptation. Drift adaptation involves updating the model either by retraining, building specialized models for different drift types, or incrementally adjusting existing models. Adaptation can be active, triggered by detected drift, or passive, with periodic updates, regardless of detection. The framework considers strategies for forgetting outdated data, either by discarding or storing it for future use. This structured approach enables models to remain accurate and responsive in non-stationary environments, particularly in regression tasks, where continuous target variables evolve over time [18].

III. CONCEPT DRIFT DETECTION AND MODEL ADAPTATION

Good Concept drift detection methods generally use a test statistic to keep tabs on the data stream and provide a measure of the similarity between old and new samples to identify the change in the concept. The similarity value was then compared with a predefined threshold to determine the drift magnitude. The generic scheme for the concept drift detection methods is shown in Figure 1. In the figure, the null hypothesis is that the test statistic would not yield a significant difference between the old and new data, that is, no concept drift is detected. If the null hypothesis is rejected, the system continues with the current learner and slides on the data stream.

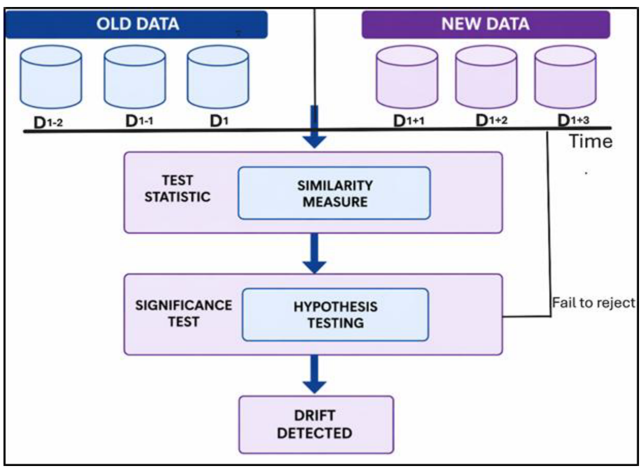

FIGURE 1. Concept drift detection.

In this study, the original dataset, the German Credit dataset, was modified by introducing different types of drifts by simulation. The first case considered to have caused concept drift was a change in the feature distribution.

Scenario 1

In this case, concept drift is introduced by increasing all the age instances by five; hence, the assumption is that the new dataset is now being considered five years later, as denoted by (1):

$$X_d = X_n + 5 \tag{1}$$

where X_d is the drifted instance and X_n is an instance in the dataset.

The drift simulation, where all age values are increased by five years, represents a temporal shift in the population, reflecting how borrower characteristics evolve over time. This mimic real-world changes in risk profiles due to aging and lifecycle effects, allowing evaluation of the model's ability to adapt to such shifts.

Scenario 2

This case was simulated by adding noise to the credit amount. This was done by increasing the credit amount using random noise from a Gaussian distribution with a mean of 0 and a standard deviation of 1000.

New credit amount = Original Credit amount + *N (0,1000)*

Here, (N (0, 1000)) represents a random variable drawn from a Gaussian (normal) distribution with a mean (μ) of 0 and standard deviation σ of 1000, as given by (2):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{2}$$

With the standard deviation, the equation becomes (3):

$$f(x) = \frac{1}{1000\sqrt{2\pi}} e^{-\frac{x^2}{2\cdot 1000^2}} \tag{3}$$

Equation (3) describes the likelihood of different values occurring in a Gaussian distribution centered around 0 with a spread determined by a standard deviation of 1000.

This represents random fluctuations and increased variability in loan sizes over time. In real-world settings, such changes may arise from factors like inflation, shifting lending policies, or evolving customer behaviour, all of which can alter the distribution of credit amounts.

Scenario 3

The original dataset has a proportion of the good and the bad represented as 70:30. This dataset was created by introducing a change in Class Distribution, in this case, credit risk, thereby reducing the proportion of the defaulted class to 20%.

The original class probabilities are denoted by ($P(C_i)$) for class (C_i). To introduce a change, a weighting factor (w_i) was applied to each class probability, as given by (4).

$$P'(C_i) = \frac{w_i \cdot P(C_i)}{\sum_j W_J \cdot P(C_j)} \tag{4}$$

where:

($P(C_i)$) is the original probability of the class C_i.

(w_i) is the weighting factor for the class (C_i)

($P'(C_i)$) is the new probability of class (C_i) after applying the weighting factor.

The denominator ensures the new probabilities sum to 1.

This reflects a shift in underlying risk prevalence. In real-world credit environments, such changes may occur due to improved economic conditions, stricter lending policies, or enhanced risk screening, all of which can lower observed default rates over time.

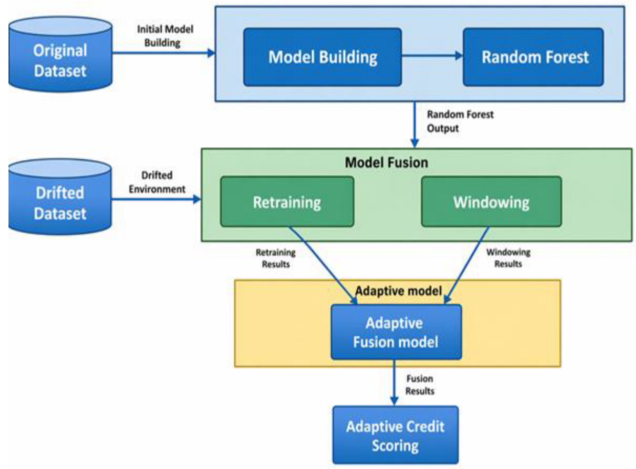

FIGURE 2. **Model adaptation architecture.**

IV. ALGORITHM DESIGN

An Adaptive method is designed to deal with feature changes in the dataset by adjusting accordingly to maintain model accuracy. The process starts by training the initial model using the vanilla models, CART, XGBoost, Naïve Bayes, and Random Forest. The machine used to run the simulations has the following specifications: Processor: 11th Gen Intel(R) Core (TM) i5-1135G7 @ 2.40GHz 2.40 GHz, Installed RAM:8.00 GB. In this vanilla testing phase, Random Forest proved to be the best performer; hence, it was used as the baseline for adaptive model development. To handle concept drift, three Random Forest models were maintained: m_0 trained on the original pre-drift data, m_1^{retain} obtained by retraining on the detected drifted data, and m_1^{window} trained using a sliding window over a recently drifted sample. For each instance, the model's output class probabilities (p_0, p_r, p_w), which are fused to produce the final prediction as given by (5):

$$jp_{final} = w_0 p_0 + w_r p_r + w_w p_w \tag{5}$$

In this case $w_0 + w_r + w_w = 1$

Figure 2 illustrates the model-adaptation architecture.

The model adaptation algorithm is detailed in Algorithm 1

V. RESULTS AND PERFORMANCE ANALYSIS

The Vanilla test results demonstrate the general performance of the algorithms under static conditions. In this particular case, there were no changing parameters or attributes in the dataset. The results showed that the Random Forest algorithm outperformed the other vanilla algorithms.

Table 1 and Figure 3 present the performance evaluation results of the vanilla model on the original dataset.

Random Forest consistently demonstrated superior results. It achieved the highest accuracy (0.77), precision (0.737), F1 score (0.680), and ROC-AUC (0.768), indicating a strong predictive capability and balanced performance in identifying true positives while minimizing false positives.

Algorithm 1 Model Adaptation

Let
D :*Original dataset*
D_{orig} :*Original data split*
D^{i}_{drift} :*Drifted versions of the dataset*, $= 1, 2 \ldots\ldots, n$
M_j *base model* $j \in \{CART, NB, RF, XGB\}$
w_t : *Sliding window of size* $W = 100$ *at time* t
Metrics $= \{Accuracy, Precison, Recall, F1, AUC\}$
Baseline model Training
For each model M_j
$M_j^{orig} = Train(D_{orig})$
$Metrics_j^{orig} = Evaluate(M_j^{orig}, D_{orig,test})$
Retraining on Drifted Data
For each drifted dataset
D^{i}_{drift} *and model* M_j
$M_j^{retrain,i} = Train(D^{i}_{drift})$
Windowed Learning
For each drifted dataset D^{i}_{drift}, time step t and model M_j
$W_t^i = D^{i}_{drift}[t - W + 1 : t]$
$M_j^{window,t,i} = Train(W_t^i)$
$Metrics_j^{window,t,i} =$
$Evaluate\ (M_j^{window,t,i},\ D^{i}_{drift}[t+1])$
Ensemble Learning
Let $\hat{y}_j^i(x)$ be the prediction from the model M_j on drifted dataset i Then
$\hat{y}^{ensemble,i}(x) =$
$mode(\{\hat{y}_j^i(x)\}_j^4 = 1)$
Adaptive Fusion
Let
m_0 : Model trained on the original data
m_r : Model retrained on drifted data
m_w : Model trained on a sliding window
Let
$p_0(x), p_r(x), p_w(x)$: Predicted class probabilities from each model
w_0, w_r, w_w : Fusion weights such that $w_0 + w_r + w_w = 1$
Then
$p_{final}(x) = w_0.p_0(x) + w_r.p_r(x) + w_w.p_w(x)$
$\hat{y}_{fusion}(x) = arg_c^{max} p_{final}(x)_c$

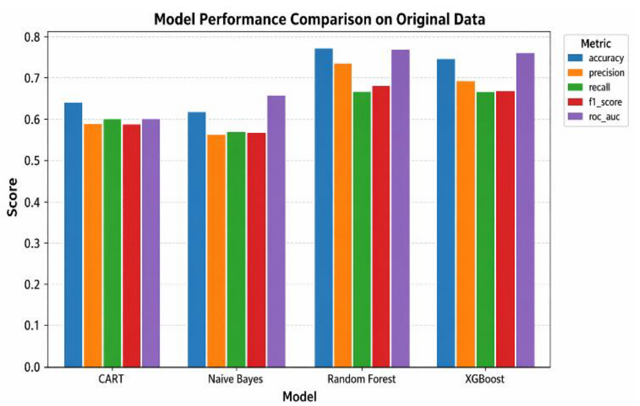

FIGURE 3. Vanilla model performance on original dataset.

XGBoost followed closely, particularly excelling in the ROC-AUC (0.760) and maintaining competitive precision and recall values. In contrast, CART and Naïve Bayes exhibited lower performance across all metrics, with Naïve Bayes slightly outperforming CART in ROC-AUC (0.662 and 0.602), but trailing in other areas. These findings suggest

TABLE 1. Vanilla model performance on original dataset.

	CART	NAIVE BAYES	RANDOM FOREST	XGBOOST
Accuracy	0.64	0.62	0.77	0.745
Precision	0.591306	0.566109	0.736796	0.689692
Recall	0.601755	0.572785	0.664382	0.666366
F1_score	0.592944	0.566951	0.680155	0.674651
ROC-AUC	0.601755	0.661617	0.7680001	0.760427

that ensemble-based methods such as Random Forest and XGBoost are more robust and effective for classification tasks.

To adapt to concept drift and evaluate the model performance, the following drift adaptation techniques were implemented: model retraining, windowing, voting ensemble, and Adaptive Fusion. Figure 4 shows the model Evaluation results on all the adaptive strategies.

FIGURE 4. ROC-AUC for adaptive retraining.

The ROC-AUC is a useful metric for detecting concept drift, particularly when dealing with class imbalance, by tracking changes in a model's ability to distinguish between classes. A significant drop in AUC over time shows that the model encounters unfamiliar patterns, indicating that the data distribution may have shifted and that retraining could be necessary. Figure 5 shows the ROC-AUC for the adaptive retraining approach.

Figure 6 shows the ROC-AUC for the adaptive Windowing approach

Figure 7 shows the ROC-AUC for the Ensemble Adaptation and Adaptive Fusion approach.

Across all three drift scenarios (drift1–drift3), the adaptive_fusion strategy delivered the strongest and most stable performance, achieving the highest values for accuracy, precision, recall, F1-score, and ROC-AUC with minimal

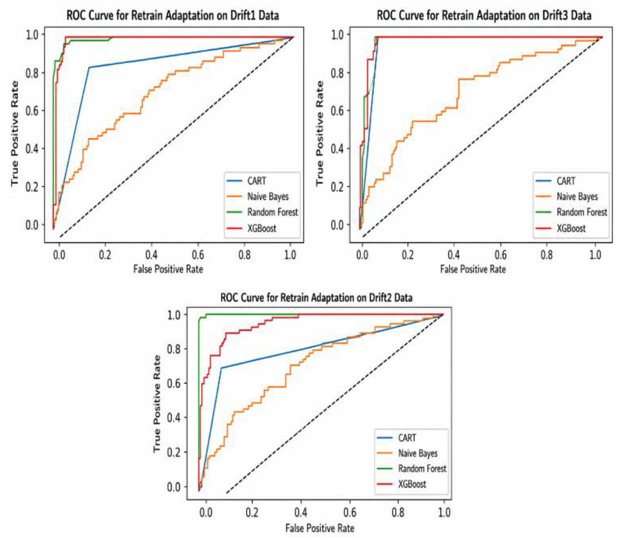

FIGURE 5. Model evaluation on adaptive strategies.

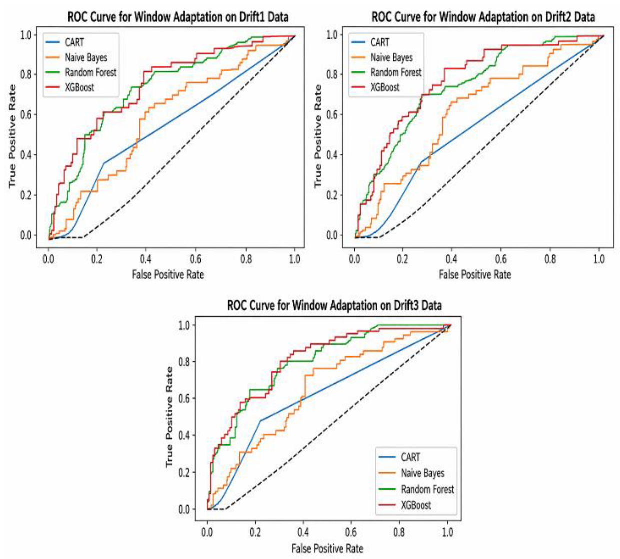

FIGURE 6. ROC-AUC for adaptive windowing.

degradation as drift occurred, indicating superior robustness to distributional shifts. The ensemble approach was consistently the second best, remaining close to adaptive_fusion but with slightly larger drops. most notably in recall; consequently, F1. Batch retraining yielded middling results: accuracy and precision were reasonable, yet recall and F1 lagged, suggesting a slower adaptation to evolving data compared with methods that combine or continuously integrate models. The sliding window method performed the worst overall, with the largest deficits in recall and F1 score. These findings support the adoption of adaptive_fusion as a default strategy for drift-prone settings.

VI. RESULTS DISCUSSION

In response to the growing challenge of concept drift in credit scoring, this study introduced a novel adaptive fusion

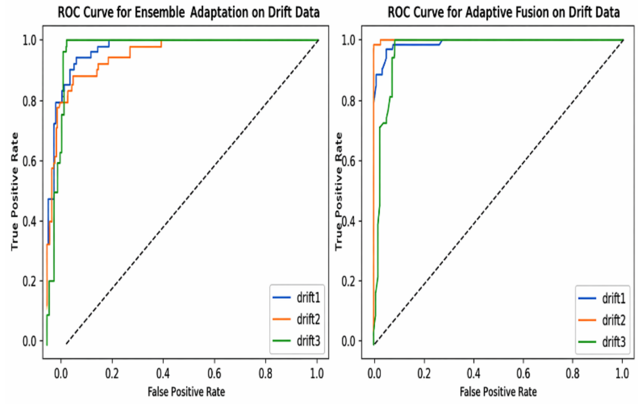

FIGURE 7. ROC-AUC for ensemble adaptation and adaptive fusion.

algorithm that dynamically integrates multiple classifiers to maintain predictive accuracy. The proposed Adaptive_ Fusion method differs from static ensembles by updating model weights dynamically rather than keeping them fixed after training. Unlike periodic retraining, it adjusts continuously based on recent performance, enabling faster and more efficient responses to concept drift. The research began by benchmarking four vanilla models: CART, Naive Bayes, Random Forest, and XGBoost, which are commonly used in financial risk modelling because of their interpretability and robustness. A German Credit Risk benchmark dataset is used. Among these, Random Forest demonstrated superior performance, consistent with the findings of Museba [13], who emphasized the effectiveness of heterogeneous ensemble models in credit scoring under dynamic conditions. To address the drift problem, four adaptive strategies, namely, retraining, windowing, ensemble, and adaptive fusion, were explored. The Adaptive Fusion outperformed all the other strategies; the Retrained Random Forest model achieved high metrics, reaffirming the value of comprehensive retraining, as supported by Krempl et al. [19], who advocated for explicit drift modelling in evolving financial datasets. The windowed Random Forest, which incrementally updates using a sliding data window, underperforms, reflecting limitations in capturing long-term behavioural shifts, a challenge also noted in adaptive online scoring models [20].

Ensemble learning based on soft voting across the four classifiers yielded good results, demonstrating its usefulness for maintaining stability without full retraining. The most significant contribution of this study is the Adaptive Fusion algorithm, which dynamically weighs and integrates model outputs based on evolving data patterns. This approach matches the retrained model in all performance metrics, offering a scalable and efficient solution to concept drift. Unlike static ensembles or retrained models, the fusion algorithm was adapted in real time.

A card-based incremental Gradient Boosting Tree (GBT) model was designed to address concept drift and data imbalance in credit card transaction streams. This model incrementally updates itself using daily reported fraudulent cards and their associated transactions, thereby maintaining a high detection performance without the need for full retraining. Compared to traditional and retrained GBT models, the incremental approach demonstrated comparable AUC scores and recall rates, while significantly reducing the training time and computational cost. Importantly, it mitigates catastrophic forgetting and eliminates the need for costly data balancing techniques. These findings suggest that similar incremental learning strategies can be effectively applied to adaptive credit scoring systems, enabling them to respond dynamically to evolving borrower behaviour and risk patterns while preserving operational efficiency [14].

In evaluating adaptive credit scoring models, the card-based incremental Gradient Boosting Tree (GBT) approach for fraud detection offers valuable insights into handling concept drift and maintaining model performance over time. Similar to adaptive fusion strategies that combine retraining and windowing, the incremental GBT model updates itself daily using transactions from reported fraudulent cards, thereby avoiding full retraining and reducing the computational cost. Both approaches aim to balance responsiveness with efficiency; however, the adaptive fusion model introduces a novel integration of dynamic windowing and selective retraining, allowing the system to adaptively respond to data shifts while preserving historical learning. This fusion enables more robust and interpretable credit scoring, particularly when combined with ensemble techniques and explainability tools. The success of incremental GBT in maintaining high AUC and recall with minimal training time supports the viability of adaptive mechanisms in financial modelling, reinforcing the relevance and innovation of adaptive fusion in credit risk prediction.

This innovation fills a critical gap in the current literature, as most adaptive credit scoring models rely on either retraining or fixed ensemble strategies. By introducing a dynamic fusion mechanism that responds to drift, this study advances the field of financial Machine Learning and offers a practical framework for real-time credit risk assessments in volatile environments.

The findings resonate well with contemporary studies, in which the designed DynED framework emphasizes the importance of diversity and dynamic selection in ensembles to effectively handle concept drift [21]. The analysis of the German Credit benchmark dataset includes several modelling approaches, such as logistic regression, discriminant analysis, tree-based methods, and random forest. The Knowledge-Maximized Ensemble (KME) is a hybrid data stream classification algorithm designed to handle various types of concept drift- abrupt, gradual, incremental, and recurrent, particularly in scenarios with limited labelled data. The KME integrates both chunk-based and online ensemble models, leveraging supervised and unsupervised knowledge to detect drift, reuse past labelled data, and recognize recurring patterns. It evaluates ensemble components based on relevance and feature space stability, thereby enabling accurate predictions through weighted classifier outputs. Experiments using Java and the MOA framework showed that KME outperforms eight state-of-the-art algorithms on synthetic and real-world datasets, offering a robust and efficient solution for dynamic data environments. On synthetic datasets, KME achieved accuracy rates ranging from 85% to 95% depending on the type of concept drift [22].

The proposed Adaptive Fusion algorithm is designed to operate alongside existing credit scoring infrastructures. It can be integrated as a complementary enhancement to existing scorecards or deployed as a parallel challenger model for performance benchmarking and validation. Outputs can be mapped to standard risk grades, ensuring compatibility with established decision workflows. The algorithm would not introduce significant scalability constraints, as model updates are performed dynamically and can be integrated within existing data processing pipelines. From a regulatory perspective, the design incorporates controlled update mechanisms, auditability and explainability features, which align with prevailing model governance requirements. As such, deployment is not expected to pose substantial compliance challenges, although standard validation and monitoring procedures remain necessary.

VII. CONCLUSION AND FUTURE WORK

This study analyzed the performance of four Machine Learning models, CART, Naive Bayes, Random Forest, and XGBoost, on the German Credit benchmark dataset under different adaptation techniques for handling concept drift. Initial evaluations on the original dataset established baseline performance, with XGBoost and Random Forest outperforming the other models, with Random Forest being the best.

Four adaptive techniques- retraining, windowed learning, ensemble learning, and Adaptive Fusion based on the Random Forest baseline were applied to test the models' ability to handle drift. Adaptive Fusion outperformed all other adaptation techniques, which implies that the adaptive integration of multiple models can achieve a high level of predictive power. This was followed by the Retrained Random, which showed high performance across all metrics, particularly excelling in recall and ROC-AUC. This shows a strong sensitivity for identifying true positives and an excellent overall classification capability.

The ensemble Model matched the retrained model in terms of accuracy, precision, recall, and F1 score, but slightly trailed in the ROC-AUC. This indicates robust performance with marginally less confidence in the ranking predictions. Visualizations, such as bar charts and ROC-AUC plots, illustrate the comparative strengths of each technique.

The adaptive Credit Scoring model accounts for fairness and bias, as historical data can reflect inequalities. By incorporating fairness-aware evaluation and transparency in adaptation, the approach aims to ensure both accuracy and equitable decision-making. Adaptive Credit Scoring enables lenders to respond more dynamically to changing borrower risk profiles, improving portfolio performance and expanding access to credit, but it also requires stronger monitoring and governance frameworks. For regulators, the use of continuously updating models raises important considerations around stability, transparency and auditability, necessitating oversight approaches that can accommodate adaptive decision systems.

Future work needs to include the use of a data stream that can be studied and analyzed to detect different kinds of concept drift the data may suffer in order to design systems to detect changes and choose a specific and different model to make predictions. This may be suitable for areas that anticipate abrupt changes that may have occurred in the past and that can be checked for in the future. The research area can be further explored as a deep learning or reinforcement problem with the application of deep learning or reinforcement algorithms to observe whether the deep learning algorithms can adapt to the changes in the original dataset.

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